Semester V												
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
MOOC	Open Elective 2 (NPTEL/SWAYAM/MOOC)	3	0	0	3	3	25	25	50	--	--	100
24HS303T	Petroleum Engineering Economics	3	0	0	3	3	25	25	50	--	--	100
24PE331T	Health Safety and Environment	3	0	0	3	3	25	25	50	--	--	100
24PE332T	Petroleum Refinery Engineering											
24PE301T	Well completion and fundamentals of Production Engineering	2	1	0	3	3	25	25	50	--	--	100
24PE302T	Well Test Analysis	2	1	0	3	3	25	25	50	--	--	100
24PE303T	Advanced Drilling Technology	3	1	0	4	4	25	25	50	--	--	100
24PE304P	Introduction to Petroleum Software Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE305P	Petroleum Engineering –I Laboratory	0	0	2	2	1	--	--	--	50	50	100
24INT251	Industrial Orientation	0	0	0	0	0	--	--	--	--	--	--
	Total	**15**	**3**	**4**	**23**	**21**						

Semester VI												
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
	Open Elective 3(From Other Department of FoET)	3	0	0	3	3	25	25	50	--	--	100
24PE333T	Data Analytics for Petroleum Engineers	2	0	0	2	2	25	25	50	--	--	100
24PE334T	Carbon Utilization and Storage											
25PE331T	Natural Gas Engineering and LNG Value Chain	3	0	0	3	3	25	25	50	--	--	100
24PE336T	Pipeline Engineering											
24PE306T	Reservoir Modelling and Simulation	2	1	0	3	3	25	25	50	--	--	100
24PE307T	Surface and Offshore Production Operations	2	1	0	3	3	25	25	50	--	--	100
24PE308T	Unconventional Hydrocarbon Energy Resources	3	0	0	3	3	25	25	50	--	--	100
24PE309P	Petroleum Engineering – II Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE310P	Petroleum Product Testing Laboratory	0	0	2	2	1	--	--	--	50	50	100
	Total	**15**	**2**	**4**	**21**	**19**						

24PE333T					Data Analytics for Petroleum Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- To develop an understanding of the big data in the oil and gas upstream industry.
- To improve skills for handling and processing data for various applications in upstream oil and gas industry.
- To improve knowledge and skills to interpret data for effective engineering operations and management.
- To develop a date based model for prediction and optimization of parameters.

Unit I: Exploratory Data Analysis **Hours: 07**

Introduction to Data Analytics in oil and gas industry, Big Data types, Characteristics of Big Data, Probability and statistics, Exploratory Data Analysis: Univariate, Bivariate, and Multivariate analysis, Parametric Distribution Models

Unit II: Prediction Models **Hours: 07**

Regression Modeling and Analysis, Simple linear regression, multiple regression, nonparametric transformation and regression; Classification and Clustering algorithms; multi-attribute regression modeling, data driven modelling; field applications and case studies, coding assignments.

Unit III: Applications In Geoscientific Applications **Hours: 06**

Application of data driven models in exploration, reservoir characterization, lithofacies interpretations and classification, petrophysical investigations using well logs, reservoir engineering, reservoir simulation and prediction optimization, physics informed machine learning models.

Unit IV: Applications In Drilling, Production and HSE Operations **Hours: 06**

Application of data science in Real-Time Monitoring, geosteering, automation, Production Engineering, Health and Safety Executive (HSE), Integrating Big Data: Simulation, Predictive Analytics, case studies to field applications.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Understand the data and correlate them with the oil and gas upstream industrial activities
CO2- Demonstrate an understanding of handling and processing big data of oil and gas upstream industry
CO3- Perform exploratory data analysis and provide statistical summary with parametric correlation
CO4- Create a statistical prediction model using multiple attributes of data
CO5- Demonstrate deep learning techniques and its application in optimization of prediction
CO6- Analyze a case study through interpretation of data for effective engineering operations and management

TEXT / REFERENCE BOOKS

1. Mishra, S., & Datta-Gupta, A. (2017). *Applied statistical modeling and data analytics: A practical guide for petroleum geosciences*. Elsevier.
2. Mohaghegh, S. D., Al-Fattah, S. M., & Popa, A. S. (2011). *Artificial intelligence and data mining applications in the E&P industry*. Society of Petroleum Engineers.
3. Mohaghegh, S. D. (2017). *Data-driven reservoir modeling: The energy industry's answer to Big Data*. Society of Petroleum Engineers.
4. Bangert, P. (2021). *Machine learning and data science in the oil and gas industry: Best practices, tools, and case studies*. Gulf Professional Publishing.
5. Pandey, Y. N., Rastogi, A., Kainkaryam, S., Bhattacharya, S., & Saputelli, L. (2020). *Machine learning in the oil and gas industry: Including geosciences, reservoir engineering, and production engineering with Python*. Apress.
6. Holdaway, K. R. (2014). *Harness oil and gas big data with analytics: Optimize exploration and production with data-driven models*. John Wiley & Sons.
7. Mount, G. (2021). *Advancing into analytics: From Excel to Python and R*. O'Reilly Media

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE334T					Carbon Utilisation and Storage					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- To introduce the fundamentals of Carbon Capture and geostorage and its role in mitigating climate change
- To develop an understanding of CO_2 capture, transportation, geological storage, monitoring, and risk assessment techniques
- To apply petroleum engineering principles to evaluate and design safe and sustainable geological carbon storage systems.

Unit I: Introduction to Carbon Capture, Utilization and Storage **Hours: 06**

Global climate change and greenhouse gas emissions; Global carbon cycle; Sources of CO_2 emissions; Net-zero emissions and energy transition; Properties of carbon dioxide; CO_2 phase behavior; CO_2 compression; Dehydration and purification; Fundamentals of CO_2 capture, Carbon capture pathways; Pre-combustion capture; Post-combustion capture; Oxy-fuel combustion; Emerging carbon capture technologies; Carbon capture efficiency and economics

Unit II: CO_2 Transportation and Utilization **Hours: 06**

CO_2 pipeline transportation; Ship transportation; Transportation network design; Safety aspects of CO_2 transportation; Introduction to CO_2 utilization; Enhanced Oil Recovery (CO_2-EOR); Industrial utilization of captured carbon

Unit III: Geological Carbon Storage **Hours: 08**

Geological storage principles; Storage site selection; Storage capacity estimation; Deep saline aquifers; Depleted oil and gas reservoirs; Unmineable coal seams; Basalt formations; Reservoir characterization; Caprock integrity; Trapping mechanisms (Structural trapping, Residual trapping, Solubility trapping, and Mineral trapping); CO_2 injectivity, Subsurface CO_2 injection, CO_2 plume migration and reservoir pressure management.

Unit IV: Monitoring, Risk Assessment and Techno-economic evaluation **Hours: 06**

Monitoring techniques: Seismic monitoring, Well logging, Pressure monitoring, Soil gas monitoring, Satellite monitoring;; Leakage pathways; Geomechanical considerations; Environmental risk assessment; Regulatory framework; Economics of geological storage; Life-cycle assessment; Techno-economic evaluation of projects, Indian and global case studies.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Explain climate change, carbon emissions, and the role of CCUS.
CO2- Describe the principles of major carbon capture technologies.
CO3- Explain CO_2 compression, transport, and utilization methods.
CO4- Evaluate geological formations suitable for CO_2 storage.
CO5- Analyze CO_2 injection, trapping, monitoring, and storage risks.
CO6- Assess technical, economic, and regulatory aspects of CCUS.

TEXT / REFERENCE BOOKS

1. Rackley, S. A. (2017). *Carbon capture and storage* (2nd ed.). Butterworth-Heinemann.
2. Smit, B., Reimer, J. A., Oldenburg, C. M., & Bourg, I. C. (2014). *Introduction to carbon capture and sequestration* (Berkeley lectures on energy). Imperial College Press.
3. Gerdes, K. F. (Ed.). (2022). *Carbon dioxide capture for storage in deep geologic formations: Results from the CO_2 Capture Project* (Vol. 5). Elsevier.
4. Ahmadian, A., Elkamel, A., & Almansoori, A. (Eds.). (2024). *Carbon capture, utilization, and storage technologies: Towards more sustainable cities* (Green Energy and Technology). Springer International Publishing.
5.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

25PE331T					Natural Gas Engineering and LNG Value Chain					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To demonstrate the concepts of Gas Processing.
- To demonstrate the Cryogenic processing of gases.
- To develop skills to plan processing required to meet market specifications economically.
- To be able to implement safe processing and handling of gases.

Unit I: Introduction **Hours: 09**

Natural Gas Properties and Natural Gas Value chain, Natural Gas Processing: Water and Hydro Carbon Liquid Separation, Dehydration, H_2S Removal and elemental Sulfur extraction, Carbon dioxide (CO_2) removal, Mercury Removal. Gas processing for pipeline transportation (Corrosion protection, limits for water, H_2S and CO_2 contents).

Unit II: Natural Gas Conditioning and Separation System **Hours: 10**

Refrigeration/ Cryogenic Process for separation/processing of Gases: Refrigeration Cycle, Cascade refrigeration processes (Multiple pure component systems and Mixed Refrigeration system), Turbo Expander and System, Advanced Refrigeration system

Unit III: Storage And Extraction of Value-Added Products From NG **Hours: 10**

Storage of Natural Gas, Extraction of components in Natural gas and /or liquids: Helium Extraction, Nitrogen Removal, Propane and Butane (LPG) Extraction, Ethane Extraction, NGL component extraction from NGL liquids by Fractionation process, NGL Extraction from Gas Mix, Introduction to use of natural gas liquid (NGL), its components for manufacture of value added products: C_2, C_3 and C_4

Unit IV: LNG **Hours: 10**

Gas processing for LNG Production (limits of Water, CO_2, H_2S and Mercury contents etc.), LNG Production process, LNG Storage, LNG Transportation & Regasification.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Understand and have a comprehensive understanding and information about the Natural Gas value chain
CO2- Understand the physical and chemical properties of Natural Gas with processing techniques
CO3- Assess techno-commercial factors involved in the development and operation of LNG regasification terminals and liquefaction processes
CO4- Understand Natural Gas transportation and storage
CO5- Enumerate global and domestic scenarios and pricing of NG
CO6- Practice employee health and safety as well as adherence to required standards for safety and operations

TEXT / REFERENCE BOOKS

1. Kumar, S. (2015). *Gas production engineering*. CRC Press.
2. Mokhatab, S., Poe, W. A., & Speight, J. G. (2019). *Handbook of natural gas transmission and processing* (4th ed.). Gulf Professional Publishing.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE336T					Pipeline Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide the basic knowledge of the pipeline operations required for transportation of oil and gas.
- Comprehensive understanding of specifications, codes, and materials of oil and gas pipeline
- Deeper insight for onshore and offshore pipeline design and fluid hydraulics
- Fundamental knowledge and understanding of commissioning, inspection, and maintenance of pipeline.

Unit I: Introduction to Oil And Gas Pipeline **Hours: 09**

Modes and comparison among different modes of transportation of petroleum products, Advantages and limitations of pipelines modes, Introduction to pipeline project, Economics and cost structure of Pipeline project, Economic pipe diameter.

Unit II: Pipeline Construction, Processes and Design Factors **Hours: 09**

Introduction to outline for Design & construction of onshore-offshore pipelines, Pipeline codes and standards, Overview of O & G field Processes, Types of Onshore/ Offshore Pipelines, Factors affecting pipeline design (External, fluid properties, pipeline parameters and Fluid flow considerations. Loop- lines

Unit III: Designing of Liquid & Gas Pipeline **Hours: 12**

Design of Liquid pipelines: Hydraulic Analysis, Relevant Pipeline Parameters, Types of fluids, Pressure Loss calculations, Maximum allowable operating Pressure, Pipeline sizing, Diameter sizing, Determination of wall Thickness, Station Spacing Pumping Power calculations, Design of Gas Pipelines: Factors affecting Gas Pipeline Design, Pressure Loss calculations, Gas pipeline Hydraulic Calculations, as Compression / Power requirement.

Unit IV: Commissioning, Inspection and Maintenance of Pipeline **Hours: 09**

Commissioning of pipeline. Pipeline Operations, Pigging, integrity assessment by Intelligent pigging and Instrumentation, Monitoring and Control Thru SCADA application, Corrosion and control/ Cathodic Protection.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Understand the basics of pipeline, roles and responsibilities of a pipeline engineer in the oil and gas Industry
CO2- Apply pipeline codes, regulations and standards in both offshore and onshore environments
CO3- Attribute pipeline terminology, various components and materials used and the fluid properties that affect pipeline transportation.
CO4- Evaluate forces acting on a pipeline system in operating conditions.
CO5- Analyse the processes involved in the prevention of corrosion and the inspection of the pipeline and its components.
CO6- Evaluate the various safety issues and practices involved in onshore and offshore pipeline operations.

TEXT / REFERENCE BOOKS

1. Alkazraji, D. (2008). *A quick guide to pipeline engineering*. Woodhead Publishing.
2. Vincent, J. (2008). *Fundamentals of pipeline engineering*. Gulf Publishing Company.
3. Menon, E. S., & Menon, P. S. (2011). *Liquid pipeline hydraulics* (2nd ed.). Trafford Publishing.
4. Antaki, G. A. (2003). *Piping and pipeline engineering: Design, construction, maintenance, integrity, and repair*. Marcel Dekker.
5. Liu, H. (2003). *Pipeline engineering*. CRC Press.
6. Lurie, M. V. (2008). *Modeling of oil product and gas pipeline transportation*. Wiley-VCH.
7. Bell, H. S. (1973). *Pipeline transportation handbook*. McGraw-Hill.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE306T					Reservoir Modelling and Simulation					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	1	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide an overview of reservoir simulation software to conceptualize the complex nature of the reservoirs
- To provide the concepts on how to create an algorithm to solve the model by applying numerical methods using the developed mathematical model & numerical model
- To explain how to apply the conceptual, mathematical and numerical skills attained on field-scale problems; and classify limitations of the conventional techniques

Unit I: Introduction **Hours: 10**

Introduction to Geological Modelling, Geostatistical Modelling & Variogram correction, classification of Models, Need for reservoir simulation, Applications of reservoir simulation, Steps of reservoir simulation, Representative Elementary Volume (REV), Property upscaling, Rock and fluid properties, PVT models, Conservation of Mass and Momentum; Continuity Equation, Equation of Motion, Darcy and Non-Darcy Flow.

Unit II: Fundamentals of Reservoir Simulation **Hours: 11**

Reservoir Discretization, Block Identification and Block Ordering, Single Phase Fluid Flow Equations in Multidimensional Domain, Flow Equations using CVFD Terminology, Various Flow Geometries, Initial and Boundary Conditions, Block-Centered and Point-Centered Grid distribution, Simulation with Block-Centered Grid, Simulation with a Point-Distributed Grid, Well Representation in Simulators, Single-Phase Flow equations for various Fluids.

Unit III: Reservoir Model Solution Algorithms **Hours: 10**

Linearization of Flow Equations, Immiscible two-phase fluid flow equations, Pseudo functions, Explicit & implicit grid system, Reservoir Model Solution Techniques: Implicit pressure and Explicit Saturation (IMPES); Sequential and Fully Implicit Method; Preview of Numerical Solution Methods: Direct & Iterative method, stability criteria, Well representation.

Unit IV: History Matching and Streamline Simulation **Hours: 08**

Model validation/verification: History Matching, Optimization algorithms in History Matching, performance prediction; Introduction to reactive transport modelling; Introduction to Streamline Simulation, comparison of Conventional / Streamline Simulation.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Apply the fundamentals of reservoir modelling and simulation
CO2- Examine and classify various models based on its application in reservoir engineering
CO3- Create Reservoir simulation models for various scenarios for visualization and implementation
CO4- Analyse and assess the required reservoir data for understanding the fluid flow in the reservoir
CO5- Validate the History matching concept for reservoir performance and production optimization
CO6- Examine the robustness of the reservoir model for performance prediction of oil and gas field reservoirs

TEXT / REFERENCE BOOKS

1. Abou-Kassem, J. H., Farouq Ali, S. M., & Islam, M. R. (2006). *Petroleum reservoir simulation*. Gulf Publishing Company.
2. Aziz, K., & Settari, A. (1979). *Petroleum reservoir simulation: A basic approach*. Applied Science Publishers.
3. Fanchi, J. R. (2006). *Principles of applied reservoir simulation* (3rd ed.). Gulf Professional Publishing.
4. Carlson, M. R. (2006). *Practical reservoir simulation*. PennWell Books.
5. Crichlow, H. B. (1977). *Modern reservoir engineering: A simulation approach*. Prentice-Hall.
6. Peaceman, D. W. (1977). *Fundamentals of numerical reservoir simulation*. Elsevier.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE307T					Surface and Offshore Production Operations					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- .Understanding the petroleum production system and its components
- Design of equipment`s used in production operations and transportation
- Analyse the use of various offshore technologies for development in shallow and deep water
- Understand the operations carried out during drilling and production in offshore environment

Unit I: Separation and Treatment of Produced Fluid **Hours: 09**

Fundamentals of surface production operations, surface facilities for liquid and gas handling, classification of separators, Components of separator, Liquid level control, design of separator, Dehydration, Desalting, Demulsification, heater treater, Effluent Treatment, Group Gathering Station, Central Tank farm, Feeder and Trunk pipeline system, pumps and compressor

Unit II: Storage and Transportation of Oil And Gas **Hours: 09**

Classification of tanks: Storage tank for Oil, storage of LPG, Underground storage, Measurement-metering of Oil and Gas, Flow assurance: problems and mitigation, Flow improver (Pour point depression and Drag reducer, heat treatment), pigging in pipe lines.

Unit III: Shallow and Deep-Water Development **Hours: 12**

Introduction to Offshore, Environmental conditions: Wave, wind and undercurrent forces. Types and classification of Offshore Platforms. Deep water development & Flow assurance strategy, floating platforms, mooring and anchoring systems, ROV and ROT for well intervention. offshore logistics, prospective new technologies.

Unit IV: Offshore Well Completion and Production System **Hours: 09**

Techniques of drilling and completing in shallow water, Techniques of drilling and completion in deep water, subsea completion, Drilling and production risers, flexible risers: configuration and laying techniques, offshore storage, Offshore production system: Fixed, floating, subsea. Smart and intelligent well completion, Oil processing in offshore and comparison with onshore. Offshore Safety and fire protection, Safety systems for offshore operations, Case history analysis and lessons learned.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Demonstrate and apply the concept of phase separators
CO2- Analyse oil and gas transport system and mitigation strategies for flow impediments
CO3- Design of surface facilities like Separators, Storage tanks and accessory equipment's
CO4- Differentiate and illustrate offshore drilling and production platforms
CO5- Analyse offshore environment and predict techniques of various operation.
CO6- Evaluate the feasibility of offshore operation in view with the offshore safety and environmental aspects

TEXT / REFERENCE BOOKS

1. Arnold, K., & Stewart, M. (1998). *Surface production operations* (Vols. 1 & 2, 2nd ed.). Gulf Professional Publishing.
2. Chilingarian, G. V., Robertson, J. O., & Kumar, S. (1987). *Surface operations in petroleum production* (Vols. 1 & 2). Elsevier. *(Vol. I published in 1987; Vol. II published in 1989.)*
3. Hughes, J. R., & Swindells, N. S. (1984). *Storage and handling of petroleum liquids: Practice and law*. Hodder Arnold.
4. Marks, A. (1978). *Petroleum storage principles*. PennWell Corporation.
5. Chakrabarti, S. K. (2005). *Handbook of offshore engineering* (Vols. 1 & 2). Elsevier.
6. Singh, S. S. P., Agarwal, J., & Mani, N. (2016). *Handbook of offshore engineering* (Vols. 1 & 2). CRC Press.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE308T					Unconventional Hydrocarbon Energy Resources					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide basic knowledge related to unconventional energy resources, their properties, and their exploitation techniques.
- To familiarize students with the unique aspects of unconventional gas and oil reservoirs, including their
  a) Resources and Economic Significance
  b) Geologic Occurrences
  c) Controls on Production Rates
  d) Drilling and Completion Practices
  e) Reservoir Management
  f) Present Activity

Unit I: Global Unconventional Energy Resources Endowment And Potential **Hours: 09**

Background and basic science about Unconventional energy resources, Comparison between formations and mode of origin, occurrences of various conventional and unconventional energy resources, Global vis-à-vis Indian unconventional energy resources and reserves, Unconventional energy resources prospects.

Unit II: Oil-Shale, Shale Gas, and Tar Sand **Hours: 10**

Oil Shale: Geological features Oil Shale formations, Chemical composition, Components of Oil Shale: Bitumen, Kerogen, Mineral matrix, Resource recovery technologies, Commercial value and Environmental impacts.
Shale gas: Definition and prospect, geological conditions for the formation of oil shale, oil shale recovery technology, ex-situ and in-situ extraction processes of shale oil, various retorting processes, processes leading to maximization of shale oil production; Shale Gas: Definition and prospect, the conditions of formation of shale gas, debate over-extraction of shale gas from the subsurface, environmental issues, American experience, Marcellus shale gas project – an example of the success story of shale gas exploitation, methods of production, hydrofracturing, the composition of fracking fluid, water management, shale gas – Indian perspective.
Tar Sand: Definition and prospect, the distinction between heavy oil and bitumen, mineralogy and properties of oil sand, elemental composition and properties of bitumen, methods of recovery of bitumen by mining and advanced in-situ processes.

Unit III: Gas Hydrate **Hours: 09**

Definition, History of Hydrate R&D, prospect, types of methane hydrate deposits, chemistry and structure of natural methane hydrate, Necessary Conditions for Methane Hydrate Formation, typical conditions of methane hydrate formation in nature vis-à vis different gas hydrate stability zones, physical properties of hydrates and ice, geology of methane hydrates, exploration for methane hydrates – geological, geochemical, and geophysical, gas hydrate – Indian perspective.

Unit IV: Coal Bed Methane **Hours: 11**

Definition and prospect, CBM, CMM, and AMM; an Overview on CBM vs. Conventional Reservoir –Gas Composition, Adsorption, Water Production, Gas Flow, Rock Physical Properties, Gas Content, Coal Rank, Gas Production. Fundamentals of Coal Geology: Genesis of Coal; Major Stratigraphic Periods of Coal Formation; Gondwana and Tertiary Coals of India; Influence of Coal Properties; Coal Chemistry – Molecular Structure, Macerals, Lithotypes, Functional Groups, Proximate Analysis, Ultimate Analysis; Significance of Rank – Definition and Measurement, Vitrinite Reflectance Measurement, Physical Properties, Volatiles Generated, Micropores; Cleat System and Natural Fracturing. Sorption: Principles of Adsorption – different types of isotherms, Langmuir Isotherm, Methane Retention; Effects of Ash and Moisture on Methane Adsorption. Decline Curves. Hydraulic Fracturing of Coal seams

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to
CO1- Understand the present global energy scenario, future needs and various unconventional hydrocarbon resources.
CO2- Analyse the Geomechanical properties of unconventional reservoirs
CO3- Outline the fundamentals of hydraulic fracturing.
CO4- Characterize the unconventional reservoirs and discuss available production methods
CO5- Apply safety and environmental features in hydraulic fracturing, gas production, and water production
CO6- Critical-thinking and problem-solving approach towards unconventional resources and recovery.

TEXT / REFERENCE BOOKS

1. Zou, C., Yang, Z., Tao, S., Yuan, X., Zhu, R., Hou, L., Yang, C., Wu, S., Lin, S., Wang, L., & Jia, J. (2013). *Unconventional petroleum geology*. Elsevier.
2. Max, M. D. (2003). *Natural gas hydrate in oceanic and permafrost environments*. Kluwer Academic Publishers.
3. Nash, K. M. (2010). *Shale gas development*. Nova Science Publishers, Inc.
4. Flores, R. M., & Moore, T. A. (2024). *Coal and coalbed gas: Future directions and opportunities*. Elsevier.
5. Dayal, A. M., & Mani, D. (2017). *Shale gas: Exploration and environmental and economic impacts*. Springer.
6. Yu, W., & Sepehrnoori, K. (2019). *Shale gas and tight oil reservoir simulation*. Gulf Professional Publishing.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE309P					Petroleum Engineering – II Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To train professional candidates capable of applying engineering principles and practices for the safe and efficient exploration, development, production, transportation and management of petroleum resources.
- To teach a student about rheological properties of any type of oil, slurry and chemical.
- To give knowledge about core flooding and hydrofracturing application in oil and gas industry.

LIST OF EXPERIMENTS

1 Determine the acid value of the given oil sample.

2 Determine the Sulphur content of the given oil sample.
3 Draw a ternary phase diagram for solubility of water benzene isopropyl alcohol (IPA) solution.
4 Determine the rheological properties of a given oil sample using Rheometer.
5 Determine the rheological properties of a given oil sample using Rheometer.
6 Determine the particle size of the given oil sample using Zetasizer Particle Size analyser.
7 Determine the Formation resistivity of the saturated rock sample.
8 Water Coning using Resistance Analogy
9 Understanding of Auto-Tensiometer
10 Understanding of Proppant Conductivity System at normal temperature conditions
11 Understanding of Proppant Conductivity System at high temperature conditions
12 Understanding of Formation Damage System with temperature conditions.
13 Understanding of Formation Damage System without temperature conditions

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to
CO1 - Determine crude oil property and its components
CO2 - Understand rheology and determine rheological properties for sample: Oil, Slurry, and Chemicals
CO3- Analyse the particle size distribution, interfacial tension, and solubility for any given fluid considering upstream and downstream applications.
CO4 -Determine reservoir physical property and productivity index using an electrical analogy.
CO5 -Evaluate proppant pack conductivity at different temperature conditions
CO6-Understanding of formation damage at different temperature conditions.

TEXT / REFERENCE BOOKS

4. Dandekar, A. Y. (2013). *Petroleum reservoir rock and fluid properties* (2nd ed.). CRC Press.
5. Begg, H. D. (1998). *Production optimization*. Oil & Gas Consultants International (OGCI) Publications.
6. Dake, L. P. (1978). *Fundamentals of reservoir engineering*. Elsevier.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**
End semester examination and Viva-voce **50 Marks**

24PE310P					Petroleum Product Testing Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To have the engineering knowledge of liquid petroleum products and their related repercussions.
- To analyse the problem related to storage, transportation and use of petroleum Products.
- To recognize the design aspects related to distillation methods.
- Properties determination of the petroleum product properties and understanding the petroleum product aromatic properties.

LIST OF EXPERIMENTS

1 Estimation of the acid number of fossil oil samples.
2 Analysis of the percentage purity of refinery product by Auto distillation apparatus.
3 Measurement of viscosity of given sample by Say Bolt viscometer.
4 Determination of viscosity of given oil sample by Red- wood Viscometer.
5 Evaluation of smoke point and Luminosity number of given kerosene sample.
6 Calculation of calorific value of coal, lignite, and oil samples by Bomb Calorimeter.
7 Determination of Flash and Fire point of refinery products by semi-automatic Prensky & Martin Apparatus.
8 Measurement of Flash and Fire point of refinery products by Cleveland open cup apparatus.
9 Evaluation of Cloud and Pour point of a given oil sample.
10 Determination of Aniline point of a given refinery products sample.
11 Estimation of Diesel Index and Cetane number of a given diesel sample.
12 Evaluation of moisture content of given liquid fuel sample using Dean and Stark apparatus.
13 Determination of Saponification value of given oil sample.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to
CO1 - Prepare themselves to understand the experiments related to midstream sector of oil and gas refineries..
CO2 - Evaluate the caloric value of crude oil sample and other fuel samples by Bomb Calorimeter.
CO3- Examine the flash and fire point of lighter and heavy petroleum products.
CO4 -Estimate the Aniline point of refinery products to confirm the aromaticity.
CO5 -Analyse the quality assurance issues as per requirements of the industrial practices.
CO6-Assess the distillation characteristics of refinery products such as gasoline, diesel, & kerosene by Auto distillation.

TEXT / REFERENCE BOOKS

7. Dandekar, A. Y. (2013). *Petroleum reservoir rock and fluid properties* (2nd ed.). CRC Press.
8. Begg, H. D. (1998). *Production optimization*. Oil & Gas Consultants International (OGCI) Publications.
9. Dake, L. P. (1978). *Fundamentals of reservoir engineering*. Elsevier.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
Evaluation based on the class performance and Laboratory book **50 Marks**
End semester examination and Viva-voce **50 Marks**